

2017 H2 Q5

Section: Our Dynamic Universe

Topic: Gravitation, redshift & stellar motion

An exoplanet (5.69×10^{27} kg) orbits a star (3.83×10^{30} kg) at 3.14×10^{11} m. Light from the star shows redshift/blueshift as the star wobbles due to the orbiting planet.

(a)(i) Compare star mass to exoplanet mass

$$\text{Ratio} = 3.83 \times 10^{30} / 5.69 \times 10^{27} \approx 673.$$

This is about 10^3 . So the star is roughly three orders of magnitude more massive.

(a)(ii) Gravitational force between star and planet

$$F = GMm / r^2.$$

$$= (6.67 \times 10^{-11} \times 3.83 \times 10^{30} \times 5.69 \times 10^{27}) / (3.14 \times 10^{11})^2$$

$$= 1.47 \times 10^{25} \text{ N}.$$

Answer: $\approx 1.5 \times 10^{25} \text{ N}$

(b)(i) Redshift when star moves at $6.60 \times 10^3 \text{ m s}^{-1}$ away

$$\text{For low speeds, redshift } z = \Delta\lambda/\lambda = v/c.$$

$$= 6.60 \times 10^3 / 3.00 \times 10^8 = 2.2 \times 10^{-5}.$$

Answer: $z \approx 2.2 \times 10^{-5}$

(b)(ii) Effect of a larger exoplanet mass

A more massive exoplanet exerts a stronger pull, so the star's circular orbit radius is greater.

Revision tips

- Orders of magnitude: compare ratio to powers of 10.
- Newton's law of gravitation: $F = GMm/r^2$.
- Redshift (for $v \ll c$): $z = v/c$.
- More massive planet → star's wobble radius increases.